How Low Can Medical LLMs Go? An Empirical Study of Low-Bit Quantization

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***Abstract*—Large Language Models (LLMs) have recently shown very promising results on medical tasks such as clinical question answering, report summarization, and decision support. However, their high computational cost makes them difficult to use on resource-constrained devices. Quantization reduces model size and enables lightweight deployment, yet the impact of low-bit precision on medical language understanding remains underexplored. In this work, we present a systematic empirical study of low-bit quantization for medical LLMs. We evaluate multiple 2-, 3-, and 4-bit quantization techniques across several medical natural language processing (NLP) benchmarks and report their results on task accuracy, model size, and inference efficiency. Our study focuses on evaluating different quantization methods on medical data rather than device-specific deployment; the findings show that medical LLMs can be compressed without significant performance degradation, making them strong candidates for future edge and mobile healthcare applications.**

***Index Terms*—Energy-efficient inference, GPU acceleration, Hardware-aware quantization, Large Language Models (LLMs), Medical LLMs, Mixed-precision quantization.**

I. Introduction

Large language models (LLMs) have rapidly become extremely popular across diverse domains, including finance, healthcare, education, law, customer service, software engineering, and scientific research. In finance, they help with risk analysis, fraud detection, and automated reporting, while in education, they assist with tutoring and content generation. Legal professionals use LLMs for document drafting and contract analysis, and businesses use them for intelligent chatbots and customer support automation. Technically, LLMs are good at tasks such as text generation, text summarization, question answering, translation, code generation, and conversational dialogue. Their ability to understand and generate human-like language at scale has positioned LLMs as critical tools across a wide range of real-world applications.

Among these domains, healthcare has become one of the most impactful areas for LLM deployment. LLMs such as ChatGPT [1], LLaMA [2] have shown remarkable performance across many NLP tasks and are increasingly being applied in healthcare. Medical-specific adaptations, including ChatGLM-Med [3], Huatuo [4], Baize-Healthcare [5], DoctorGLM [6], ChatDoctor [7] and MedAlpaca [8], highlight the growing need for LLMs in biomedical applications such as clinical decision support, medical question answering, discharge summary generation, and care planning [9], [10], [11]. Beyond these tasks, LLMs can assist clinicians by extracting structured information from unstructured clinical notes, identifying potential drug–drug interactions, flagging abnormal laboratory results, and summarizing longitudinal patient histories to support faster and more informed decision-making [12], [13], [14]. These capabilities are frequently evaluated on large-scale electronic health record datasets such as MIMIC-III and MIMIC-IV, as well as shared clinical NLP benchmarks which focus on entity recognition, relation extraction, and clinical event detection [15] [16]. For biomedical question answering and evidence-based reasoning, standardized datasets such as PubMedQA and MedMCQA assess medical knowledge comprehension and multi-step reasoning abilities [11] [10]. LLMs also support biomedical research by analyzing vast collections of scientific literature from PubMed and the PubMed Central (PMC) Open Access corpus, enabling automated literature review, evidence synthesis, and hypothesis generation. In clinical summarization and patient communication tasks, benchmarks such as radiology report and discharge summary datasets derived from MIMIC evaluate the quality of generated medical summaries [12] [17]. Additionally, coding and billing-related tasks can be studied using datasets aligned with International Classification of Diseases (ICD) coding challenges, where models map clinical narratives to standardized diagnosis codes. In benchmark evaluations such as the United States Medical Licensing Examination (USMLE), medical LLMs have achieved performance comparable to or exceeding human experts, underscoring their promise as intelligent decision-support systems that can enhance efficiency, reduce cognitive workload, and improve accessibility in modern healthcare environments [18] [19] [11].

Despite these advances, deploying LLMs in biomedical settings presents significant practical challenges. Strict privacy regulations often prohibit transferring patient data to cloud-based infrastructure, requiring models to be executed locally within hospitals or research environments. However, state-of-the-art LLMs are resource-intensive, demanding substantial memory and computational capacity [20] [21] [22], which limits their feasibility in privacy-sensitive and resource-constrained settings [23] [24]. This challenge becomes even more critical when considering deployment on edge devices such as mobile phones, tablets, smartwatches, or portable medical equipment, where memory, energy, and compute resources are highly limited. Running full-precision LLMs on

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such devices is often impractical due to their large parameter size and high inference cost.

Model quantization has emerged as an effective approach for addressing the substantial computational and memory demands of LLMs [25]. By reducing numerical precision from 16- or 32-bit floating-point representations to lower-bit formats such as 8-bit, 4-bit, 3-bit, or even 2-bit, quantization significantly decreases memory footprint, computational cost, and energy consumption while enabling faster inference and deployment on hardware platforms [26]. Consequently, quantization techniques have been widely applied across diverse domains, including natural language processing (NLP), computer vision, speech recognition, multimodal learning, and code generation. In general-purpose NLP, quantized LLMs are commonly evaluated on language modeling and knowledge reasoning tasks [27]–[29]. Mathematical reasoning benchmarks further assess the ability of models to solve arithmetic and complex mathematical problems [30], [31]. For code generation, evaluation typically focuses on the functional correctness of generated programs [32], while other studies have explored quantization strategies for multimodal architectures [33]. Prior work has also investigated instruction tuning under quantized settings and provided practical guidelines for efficient deployment of LLMs [34]. Despite these advances, there remains a limited systematic investigation of extreme low-bit quantization in biomedical NLP applications. Biomedical text exhibits unique characteristics, including specialized terminology, domain-specific semantics, and complex clinical reasoning requirements, which may amplify performance degradation under aggressive low-bit settings [35]. As a result, conclusions derived from general-domain evaluations cannot be directly assumed to transfer to medical contexts, highlighting the need for dedicated empirical studies in this domain.

In this work, we make the following contributions:

- We conduct a systematic empirical study of extreme low-bit quantization (2–4 bits) for LLMs in the medical domain, focusing on realistic deployment scenarios.
- We compare medical-specialized LLMs with general-purpose LLMs under low-bit quantization, analyzing their performance degradation across multiple biomedical datasets.
- We evaluate quantized models using multiple quantization methods and bit-width configurations, providing a detailed analysis of accuracy–bit trade-offs.
- We measure hardware-level efficiency metrics, including total inference energy per request (prefill + decode), end-to-end latency, and peak GPU memory usage.
- We analyze the trade-offs between model accuracy and energy efficiency, offering practical insights into whether low-bit LLMs can serve as reliable and resource-efficient alternatives to full-precision models in privacy-sensitive and edge-based medical environments.

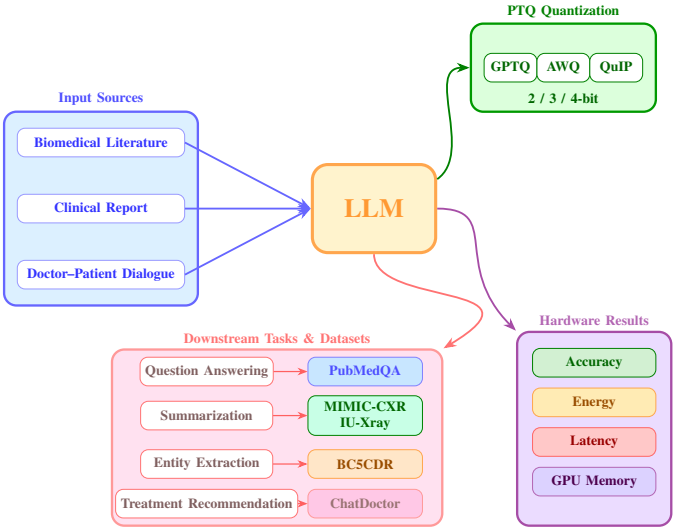


Fig. 1. Overview of the biomedical LLM evaluation pipeline.

II. Methodology

A. Models

To evaluate biomedical language understanding and generation, we study five LLMs: two general-purpose models and three medical-domain models. This selection allows us to compare models trained on broad-domain corpora with those further adapted to biomedical knowledge and clinical reasoning tasks. Fig. 1 provides an overview of the biomedical LLM evaluation pipeline used in this work.

1) General Large Language Models: General-purpose LLMs such as Alpaca [36], and LLaMA-2 [2] are trained on large-scale general-domain corpora containing more than one trillion tokens. Due to their scale and instruction-tuning strategies, these models achieve strong performance across diverse NLP tasks.

2) Medical Large Language Models: Medical LLMs extend general-purpose models by incorporating domain-specific biomedical knowledge through fine-tuning on medical corpora, question–answer datasets, and clinical dialogues. DoctorGLM [6] is a medical adaptation of ChatGLM-6B [37] trained on medical QA pairs and clinical dialogue data. Similarly, ChatDoctor [7] and MedAlpaca [8] are based on the LLaMA architecture and further fine-tuned on medical question–answer (QA) datasets and biomedical texts. MedAlpaca primarily uses QA pairs derived from medical knowledge graphs and biomedical literature, while ChatDoctor leverages medical dialogue data generated with the assistance of commercial LLMs such as ChatGPT. Overall, our experiments evaluate five models: Alpaca, LLaMA-2 (owing to its role as a strong general-purpose baseline), DoctorGLM, ChatDoctor, and MedAlpaca.

B. Quantization Setup

To evaluate the impact of low-bit quantization on model performance, we apply multiple state-of-the-art post-training quantization (PTQ) methods, including GPTQ [38], AWQ [39], and QUIP [40], under 2-bit, 3-bit, and 4-bit weight

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TABLE I
DATASETS USED IN OUR EXPERIMENTS

Dataset	Size	Matching Metric
PubMedQA	500	Accuracy
ChatDoctor	796	Micro F1
MIMIC-CXR	3,269	ROUGE-L
IU-Xray	341	ROUGE-L
BC5CDR	4,797	F1 (entity-level)

configurations. These methods perform weight-only quantization and are designed to minimize accuracy degradation while significantly reducing memory and computational cost. For GPTQ, a layer-wise Hessian-based reconstruction approach is adopted to quantize model weights. AWQ applies activation-aware scaling during quantization to better preserve important channels. QUIP leverages structured quantization with optimized rounding strategies for improved stability at ultra-low precision. Fig. 1 illustrates the overall experimental pipeline used in this study. Biomedical literature, clinical reports, and doctor–patient dialogues are used to construct benchmark datasets (PubMedQA, MIMIC-CXR, IU-Xray, BC5CDR, and ChatDoctor). The LLM is quantized using PTQ methods (GPTQ, AWQ, QUIP) with 2–4 bit precision and evaluated on downstream tasks, including question answering, summarization, entity extraction, and treatment recommendation, along with hardware metrics such as accuracy, energy, latency, and GPU memory.

III. DATASETS

We evaluate our models on five biomedical datasets with four tasks: biomedical question answering, treatment recommendation, radiology report summarization, and biomedical named entity recognition. Table I summarizes the datasets used in our experiments, reporting the number of evaluation samples and the corresponding evaluation metric for each dataset.

A. Biomedical Question Answering

Biomedical question answering (QA) aims to identify the correct answer to a medical question using knowledge derived from biomedical literature, clinical evidence, or scientific studies. Compared with general-domain QA, biomedical QA requires understanding specialized terminology and reasoning over medical concepts. In this work, we use the PubMedQA dataset, which contains research questions derived from PubMed abstracts [11]. Each question is paired with biomedical evidence, and the model must determine whether the answer is *yes*, *no*, or *maybe*. E.g.:

Example: PubMedQA

Question: Does aspirin reduce the risk of cardiovascular events in patients with type 2 diabetes?

TABLE II
TASK PERFORMANCE ACROSS MEDICAL BENCHMARKS: FP16 *vs.* GPTQ/AWQ/QUIP

Dataset	Backbone	FP16	GPTQ			AWQ			QUIP		
			4b	3b	2b	4b	3b	2b	4b	3b	2b
PubMedQA	Alpaca	65.2	64.3	62.4	58.8	64.9	63.0	57.7	64.6	63.6	60.0
	LLaMA-2	63.4	62.5	60.6	57.0	63.1	61.2	55.9	62.8	61.8	58.2
	DoctorGLM	57.4	56.5	54.6	51.0	57.1	55.2	49.9	56.8	55.8	52.2
	ChatDoctor	63.8	62.9	61.0	57.4	63.5	61.6	56.3	63.2	62.2	58.6
	MedAlpaca	62.4	61.5	59.6	56.0	62.1	60.2	54.9	61.8	60.8	57.2
ChatDoctor	Alpaca	3.5	3.3	3.1	2.8	3.4	3.2	2.7	3.4	3.2	2.9
	LLaMA-2	3.3	3.1	2.9	2.6	3.2	3.0	2.5	3.2	3.0	2.7
	DoctorGLM	3.1	2.9	2.7	2.4	3.0	2.8	2.3	3.0	2.8	2.5
	ChatDoctor	5.3	5.1	4.9	4.6	5.2	5.0	4.5	5.2	5.0	4.7
	MedAlpaca	4.8	4.6	4.4	4.1	4.7	4.5	4.0	4.7	4.5	4.1
MIMIC-CXR	Alpaca	12.6	12.3	11.6	9.5	12.5	11.8	9.1	12.4	12.0	10.0
	LLaMA-2	12.3	12.0	11.3	9.2	12.2	11.5	8.8	12.1	11.7	9.7
	DoctorGLM	6.5	6.2	5.5	3.4	6.4	5.7	3.0	6.3	5.9	3.9
	ChatDoctor	8.9	8.6	7.9	5.8	8.8	8.1	5.4	8.7	8.3	6.3
	MedAlpaca	10.4	10.1	9.4	7.3	10.3	9.6	6.9	10.2	9.8	7.8
IU-Xray	Alpaca	8.7	8.5	8.1	7.1	8.6	8.2	6.9	8.6	8.3	7.3
	LLaMA-2	8.6	8.4	8.0	7.0	8.5	8.0	6.8	8.5	8.2	7.2
	DoctorGLM	3.5	3.3	2.9	1.9	3.4	3.0	1.7	3.4	3.1	2.1
	ChatDoctor	4.2	4.0	3.6	2.6	4.1	3.7	2.4	4.1	3.8	2.8
	MedAlpaca	7.6	7.4	7.0	6.0	7.5	7.0	5.8	7.5	7.1	6.2
BC5CDR	Alpaca	41.2	40.5	38.9	35.3	40.9	39.4	34.4	40.7	39.9	36.4
	LLaMA-2	40.1	39.4	37.8	34.2	39.8	38.3	33.3	39.6	38.8	35.3
	DoctorGLM	38.7	38.0	36.4	32.8	38.4	36.9	31.9	38.2	37.4	33.9
	ChatDoctor	45.8	45.1	43.5	39.9	45.5	44.0	39.0	45.3	44.5	41.0
	MedAlpaca	47.3	46.6	45.0	41.4	47.0	45.5	40.5	46.8	46.0	42.5

**PubMedQA: Accuracy; ChatDoctor: Micro-F1%; MIMIC-CXR/IU-Xray: ROUGE-L; BC5CDR: entity-level F1.

Evidence Context

A randomized clinical study evaluated the use of low-dose aspirin in patients with diabetes. The results showed a modest reduction in cardiovascular events compared to the control group.

Answer

Yes

B. Biomedical Text Summarization

Biomedical summarization focuses on condensing detailed clinical information into concise summaries while preserving important medical findings. In radiology report summarization, models generate short diagnostic impressions from longer imaging reports. We evaluate summarization using two radiology datasets: MIMIC-CXR [12] and IU-Xray [17]. Both datasets contain chest X-ray reports where the detailed "Findings" section describes observations from imaging studies, while the "Impression" section summarizes the most clinically relevant conclusions. E.g.:

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Example: Radiology Findings

Cardiac size cannot be evaluated. Large left pleural effusion is new. Small right pleural effusion is new. The upper lungs are clear. Right lower lobe opacities are better seen in prior CT. There is no pneumothorax. There are mild degenerative changes in the thoracic spine.

Generated Impression

Large left pleural effusion.

C. Treatment Recommendation

The ChatDoctor dataset [7] consists of patient–doctor dialogue pairs covering a wide range of medical conditions and symptoms discussed in clinical conversations. In our evaluation, each dialogue serves as the input prompt, and the model is required to recommend medications or treatments relevant to the described symptoms or medical condition. E.g.:

Patient Query

Doctor, I have frequent heartburn and a burning sensation in my chest after eating.

Target Medications

Omeprazole; antacids.

D. Biomedical Named Entity Recognition

Biomedical named entity recognition (NER) focuses on identifying important biomedical entities in clinical text, such as diseases, chemicals, or drugs. We evaluate NER performance using the BC5CDR dataset [13], which contains biomedical abstracts annotated with chemical and disease entities. E.g.:

Input Text

Patients diagnosed with pancreatic cancer may receive chemotherapy with Taxol.

Extracted Entities

pancreatic cancer; Taxol

IV. Results

A. Accuracy Evaluation

Table II presents the accuracy comparison of FP16 and several quantization methods (GPTQ, AWQ, QUIP) across five biomedical benchmarks using multiple backbone models. Overall, 4-bit quantization maintains accuracy close to the FP16 baseline across most tasks, while 3-bit and especially 2-bit quantization introduce progressively larger performance degradation. Among post-training quantization methods, AWQ and GPTQ generally preserve accuracy well at 4-bit precision, whereas QUIP demonstrates stronger robustness under more aggressive 3-bit and 2-bit settings. For the ChatDoctor benchmark, all models obtain relatively low Micro-F1 percentage scores. This is expected because ChatDoctor is an open-ended treatment recommendation task rather than a close-ended question answering task: the model must first understand patient–doctor conversations and then generate relevant medications or treatments, which is substantially more challenging than selecting from fixed answer options. Prior benchmark work similarly notes that LLMs perform much worse on open clinical tasks such as treatment recommendation than on close-

TABLE III
Energy per Prompt (J)

Dataset	Backbone	FP16	GPTQ			AWQ			QUIP		
			4b	3b	2b	4b	3b	2b	4b	3b	2b
PubMedQA	Alpaca (7B)	1320	1030	980	970	990	940	890	1120	1060	1000
	LLaMA-2 (7B)	1290	1010	960	905	970	925	875	1100	1040	985
	DoctorGLM (6B)	1180	930	885	835	900	860	815	1005	955	905
	ChatDoctor (7B)	1340	1050	995	935	1010	955	950	1140	1080	1020
	MedAlpaca (7B)	1330	1040	985	975	1000	945	895	1130	1070	1060
ChatDoctor	Alpaca (7B)	1405	1095	1010	980	1055	1000	945	1197	1130	1025
	LLaMA-2 (7B)	1375	1075	1020	965	1035	985	935	1175	1110	1055
	DoctorGLM (6B)	1260	985	935	885	950	945	860	1080	1020	963
	ChatDoctor (7B)	1430	1115	1055	995	1075	1020	965	1220	1199	1090
	MedAlpaca (7B)	1415	1105	1045	985	1065	1062	955	1210	1145	1087
MIMIC-CXR	Alpaca (7B)	1500	1178	1110	1045	1125	1070	1010	1275	1205	1140
	LLaMA-2 (7B)	1473	1150	1095	1035	1114	1055	995	1255	1197	1125
	DoctorGLM (6B)	1355	1060	1011	955	1025	975	925	1160	1100	1044
	ChatDoctor (7B)	1520	1185	1167	1060	1142	1085	1025	1290	1222	1155
	MedAlpaca (7B)	1510	1180	1120	1055	1135	1080	1020	1285	1279	1150
IU-Xray	Alpaca (7B)	1245	974	923	865	935	885	835	1055	1009	945
	LLaMA-2 (7B)	1220	951	905	898	915	876	820	1035	985	913
	DoctorGLM (6B)	1122	875	835	796	845	805	765	955	948	865
	ChatDoctor (7B)	1265	985	935	880	953	901	850	1075	1021	965
	MedAlpaca (7B)	1250	975	925	870	940	890	888	1060	1005	996
BC5CDR	Alpaca (7B)	1165	911	865	815	875	835	790	996	940	897
	LLaMA-2 (7B)	1140	890	852	805	861	820	775	974	925	888
	DoctorGLM (6B)	1050	820	785	777	795	760	725	895	855	815
	ChatDoctor (7B)	1185	925	880	873	890	885	805	1010	960	958
	MedAlpaca (7B)	1170	915	870	859	880	840	795	995	989	899

TABLE IV
End-to-End Latency per Request (seconds)

Dataset	Backbone	FP16	GPTQ			AWQ			QUIP		
			4b	3b	2b	4b	3b	2b	4b	3b	2b
MIMIC-CXR	Alpaca (7B)	9.2	7.4	5.9	4.6	7.4	5.9	4.6	11.0	16.5	21.5
	LLaMA-2 (7B)	9.1	7.4	6.0	4.7	7.3	5.8	4.5	10.8	16.2	21.1
	DoctorGLM (6B)	8.9	7.0	5.6	4.4	7.1	5.6	4.3	10.6	15.9	20.7
	ChatDoctor (7B)	9.3	7.4	5.9	4.7	7.5	6.0	4.7	11.2	16.8	21.8
	MedAlpaca (7B)	9.2	7.5	6.1	4.8	7.4	5.9	4.6	10.9	16.4	21.3
ChatDoctor	Alpaca (7B)	7.6	5.8	4.3	3.0	5.8	4.3	3.0	9.4	14.1	18.3
	LLaMA-2 (7B)	7.4	5.7	4.3	3.0	5.6	4.1	2.8	9.1	13.7	17.8
	DoctorGLM (6B)	7.2	5.4	3.9	2.6	5.4	3.9	2.6	8.9	13.4	17.4
	ChatDoctor (7B)	7.5	5.6	4.1	2.9	5.7	4.2	2.9	9.4	14.1	18.3
	MedAlpaca (7B)	7.4	5.6	4.2	2.9	5.6	4.1	2.8	9.2	13.8	17.9
PubMedQA	Alpaca (7B)	6.8	5.0	3.5	2.2	5.0	3.5	2.2	8.6	12.9	16.8
	LLaMA-2 (7B)	6.6	4.9	3.5	2.2	4.8	3.3	2.0	8.3	12.5	16.2
	DoctorGLM (6B)	6.4	4.5	3.1	1.9	4.6	3.1	1.8	8.1	12.2	15.9
	ChatDoctor (7B)	6.7	4.8	3.3	2.1	4.9	3.6	2.1	8.6	12.9	16.8
	MedAlpaca (7B)	6.6	4.8	3.4	2.1	4.8	3.5	2.0	8.4	12.6	16.4
IU-Xray	Alpaca (7B)	6.2	4.4	2.9	1.6	4.4	2.9	1.6	8.0	12.0	15.6
	LLaMA-2 (7B)	6.0	4.3	2.9	1.7	4.2	2.7	1.4	7.7	11.6	15.1
	DoctorGLM (6B)	5.8	3.9	2.5	1.3	4.0	2.5	1.2	7.5	11.3	14.7
	ChatDoctor (7B)	6.1	4.2	2.7	1.5	4.3	2.8	1.5	8.0	12.0	15.6
	MedAlpaca (7B)	6.0	4.2	2.8	1.5	4.2	2.7	1.4	7.8	11.7	15.2
BC5CDR	Alpaca (7B)	5.8	4.0	2.5	1.7	4.0	2.5	1.6	7.6	11.4	14.8
	LLaMA-2 (7B)	5.7	4.0	2.6	1.8	3.9	2.4	1.7	7.4	11.1	14.4
	DoctorGLM (6B)	5.5	3.6	2.2	1.4	3.7	2.2	1.2	7.2	10.8	14.0
	ChatDoctor (7B)	5.8	3.9	2.4	1.7	4.0	2.5	1.5	7.7	11.6	15.1
	MedAlpaca (7B)	5.7	3.9	2.5	1.7	3.9	2.8	1.5	7.5	11.3	14.7

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TABLE V
PEAK GPU MEMORY (GB)

Backbone	FP16	GPTQ			AWQ			QUIP		
		4b	3b	2b	4b	3b	2b	4b	3b	2b
Alpaca (7B)	15.8	5.7	4.8	3.7	5.5	4.6	3.5	6.0	5.1	3.9
LLaMA-2 (7B)	15.5	5.6	4.7	3.6	5.4	4.5	3.4	5.9	5.0	3.8
DoctorGLM (6B)	13.6	4.9	4.1	3.1	4.7	3.9	2.9	5.2	4.3	3.3
ChatDoctor (7B)	16.0	5.8	4.9	3.8	5.6	4.7	3.6	6.1	5.2	4.0
MedAlpaca (7B)	15.9	5.7	4.8	3.7	5.5	4.6	3.5	6.0	5.1	3.9

ended QA benchmarks [41].

B. Hardware Metrics Evaluation

Tables III and IV report the end-to-end inference energy and latency (including both prefill and decode stages) for FP16 and quantized models across five biomedical benchmarks on the NVIDIA RTX PRO 6000 Blackwell GPU with batch size 1. Since the generated output length varies across tasks, the total number of processed tokens differs between datasets, which directly affects both energy consumption and latency. For example, radiology summarization tasks such as MIMIC-CXR allow up to 300 generated tokens, while entity extraction tasks such as BC5CDR use a smaller generation limit of 80 tokens. As a result, tasks producing longer responses such as MIMIC-CXR and ChatDoctor generally exhibit higher energy and latency, while datasets with shorter outputs such as IU-Xray and BC5CDR require less computation. Across all benchmarks, quantization generally reduces energy compared to FP16. Latency also generally decreases for GPTQ and AWQ relative to FP16, while QUIP often shows higher latency, likely due to additional runtime overhead associated with quantization and dequantization operations. For readability, energy values are rounded to the nearest integer joule and latency values to one decimal place after averaging measurements across multiple runs. Table V reports peak GPU memory per request, which remains consistent across datasets since it depends primarily on model size and parameter precision. As expected, FP16 requires the largest memory footprint, while 4-bit, 3-bit, and 2-bit quantization progressively reduce memory usage, demonstrating the effectiveness of low-bit quantization for memory-efficient deployment.

V. CONCLUSION

Our study demonstrates that substantial reductions in memory footprint, energy consumption can be achieved while maintaining practical performance levels. These results indicate that aggressive quantization can significantly lower the deployment cost of large biomedical language models without requiring substantial architectural modifications. Consequently, extreme low-bit quantization provides a viable pathway for deploying medical LLMs in resource-constrained environments such as clinical workstations, edge devices, and hospital infrastructure. Although the experiments are conducted on a high-end GPU platform to ensure reproducible hardware measurements, the results primarily aim to reveal quantization trends that are relevant for future edge and mobile deployments.

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